Overview

Title
Introduction into Computer Science (for non Informatics studies, TUM BWL) (IN8005)

Number
240905438

Persons involved
Lecturer (Assistant)
Groh, Georg

Type
lecture (VO)

Semester weekly hours
2

ECTS credits
-

Course language/s
English

Offered in
Winter semester 2024/25

Organisation
Informatics 2 - Chair of Formal Languages, Compiler Construction, Software Construction (Prof. Seidl)

Description

Support

Help

Content
Content:
The module IN8003 is concerned with topics such as:
• Database Management Systems,.ER models, Relational Algebra, SQL
• Java as a programming language:
o basic constructs of imperative programming (if, while, for, Arrays etc.)
o object-oriented programming (inheritance, Interfaces, Polymorphism etc.)
o basics of Exception Handling and Generics
o code conventions
o Java class library
• Basics of Visual Basic for Applications
• Basic algorithms and data structures:
o algorithm concept, complexity
o data structures for sequences (arrays, doubly linked lists, stacks & queues)
o recursion
o hashing (chaining, probing)
o searching (binary search, balanced search trees)

Previous knowledge expected
Recommended requirements are Mathematics modules of the first year of the TUM-BWL bachelor's program as well as the module WI000275 'Management Science'. If you did not attend these, it is not a big problem.

Objective
Upon successful completion of the module, participants understand important foundations, concepts and ways of thinking of Computer Science, in particular object-oriented programming, databases and SQL, and basic algorithms and data structures, have an overview over these topics and be able use them for the development of own programs with a link to a database in a basic way.

Teaching and learning method
interactive lecture

Lecture and practical tutorial assignments. A central tutorial deepens the understanding of the concepts introduced in the lecture using example assignments in regard to being able to solve given problems. In the tutorials, the students solve basic assignments under intensive supervision, which contributes to providing them with the basic skills in programming, in order to be able to apply the knowledge acquired by self-study of the accompanying materials of lecture and central tutorial for autonomously solving the programming assignments of the homework. During the second half of the semester, the students work on a small practical project, which aims at deepening the connected understanding of the desired learning outcomes. Programming aspects of this project are distributed over tutorial and homework assignments and are aligned with the topics of the respective week.

Course Criteria and Registration
(free registration via tum-online)

Further information
Recommended Reading
Chapters from textbooks, which are closely associated with the module content and are provided to the students online.

Online information
e-learning course (moodle)

Categories

Support

Dates and Groups

Help

Standardgruppe

Lecturer Groh, Georg

Monday , 15:00 - 16:30 **from 21.10.2024 to 03.02.2025**
0980, Audimax (0509.EG.980)

Series

Status within Curriculum

My Studies

Management and Technology (Main location: Heilbronn) - Bachelor of Science (1630 17 247) , 20201 (offered until 30.09.2050)

[20201] Bachelorstudiengang Management and Technology (Heilbronn)
Specialization in Technology
Specialization in Technology: Digital Technologies
Required Modules: Digital Technologies
[IN8005] Introduction into Computer Science (for non informatics studies)
Introduction into Computer Science (for non Informatics studies, TUM BWL) (IN8005)

Subject type Compulsory elective subject | **Recommended** 3rd semester | **Preconditions** None

Go to my programme

Other Degree Programmes

Equivalent courses

Winter semester 2024/25

240905438
Introduction into Computer Science (for non Informatics studies, TUM BWL) (IN8005)
VO | **2** SWS | **-** ECTS | 2024 W
Lecturer
Groh, Georg

Further

Support

Help

Support
Help